

Industry Value-Chain & the Margin-Distortion Anomaly

Conform ~8 of Damodaran's industry datasets onto one `dim_industry` spine in DuckDB, test one honest cross-dataset thesis, and export a Supabase-ready Postgres database. The value is the **join + the thesis + visible honesty** — not the plumbing.

DIFFICULTY

Advanced

Entity resolution + accounting literacy

EFFORT

~55–80 h

Reuses the portfolio plumbing

UNIVERSE

94 industries

Jan-2026, TTM ≈ Q3 2025

TABLES

3 dims · 8 facts

+ wide mart, in DuckDB
→ Postgres

DATA

★★★★☆

Clean values, dirty join key

HEADLINE FINDING

EV/Sales tracks Damodaran's R&D/lease-**adjusted** operating margin better than the **reported** margin (R^2 0.425 vs 0.406) across 84 non-financial industries — the market looks partly through R&D expensing. The adjustment reranks only 1/84 by ≥ 5 places ($p = 0.998$): the distortion is concentrated in the high-R&D tail.

■ THE THESIS (ONE CLAIM)

Reported margins **misprice the R&D-heavy tail** in cross-industry screening — and the equity market already prices the adjusted figure.

So what: don't screen software / semis / pharma on reported operating margin or P/E. **Non-circular by design:** R&D-adjusted margins are Damodaran's own construction, so we never regress them on R&D intensity — the empirical test points at a market outcome (EV/Sales) instead. A lint blocks the circular pairing.

■ PIPELINE (RAW → STAGING → MART → POSTGRES)

Ingest

6 canonical .xls + hash

→

Parse

dynamic header, NA→NULL

→

Resolve

alias crosswalk, fail-loud

→

Warehouse

DuckDB star schema

→

Export

Supabase SQL + RLS

■ ENGINEERING SIGNALS

- **Join on a surrogate id, never the raw string** — catches Damodaran's real Healthcare typo; fails loudly on any unresolved name.
- **Vintage dimension** — single snapshot → panel by design; ASC 842 / TCJA §174 break flags.
- **15 tests** + firm-count anchor (5,994) + anti-circularity lint, all green.
- **Right-sized** — 8 facts, not 30; proportionality over gold-plating.

■ WHAT IT CAN & CANNOT SHOW

- Value-weighted industry **aggregates** ($n \leq 94$) — the "industry portfolio", not a typical firm.
- **Association, not causation** — ecological fallacy, single snapshot.
- Capitalized R&D is Damodaran's **estimate**, not GAAP.
- US-only; Total Market rows stripped before every statistic.